Coding & Solution

SMARTBRIDGE

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y
Programming Language(s)	JavaScript (Node.js runtime environment v16+)
Framework(s) Used	Express.js (REST API framework), Mongoose ODM
Key Features Implemented	JWT User Registration, Login & Profile Retrieval. Full Workout CRUD Management (Create, Read, Search, Update, Delete). Google Gemini AI Workout Recommendations & Performance Insights. Centralized JSON Error Handling & Schema Validations.
Pending / Incomplete Features	None. All planned backend REST endpoints and AI features are 100% operational.
Setup / Run Instructions	1. Clone repository and navigate into project directory: cd Code Files. 2. Install dependencies: npm install. 3. Configure .env file with PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY. 4. Start server: npm start. 5. Execute API tests via Thunder Client / Postman at http://localhost:5000.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

6	Code is committed to a version control repository	Yes

Additional Notes / Comments: Application verified successfully using Thunder Client. API endpoints handle database operations and Gemini AI requests within target response times.